HOLOCENE DIVERSITY TRENDS

A COMPARATIVE ANALYSIS OF METHODOLOGICAL FRAMEWORKS FOR RECONSTRUCTING CONTINENTAL DIVERSITY TRENDS OF VEGETATION : CASE OF THE NORTHERN HEMISPHERE DURING THE HOLOCENE

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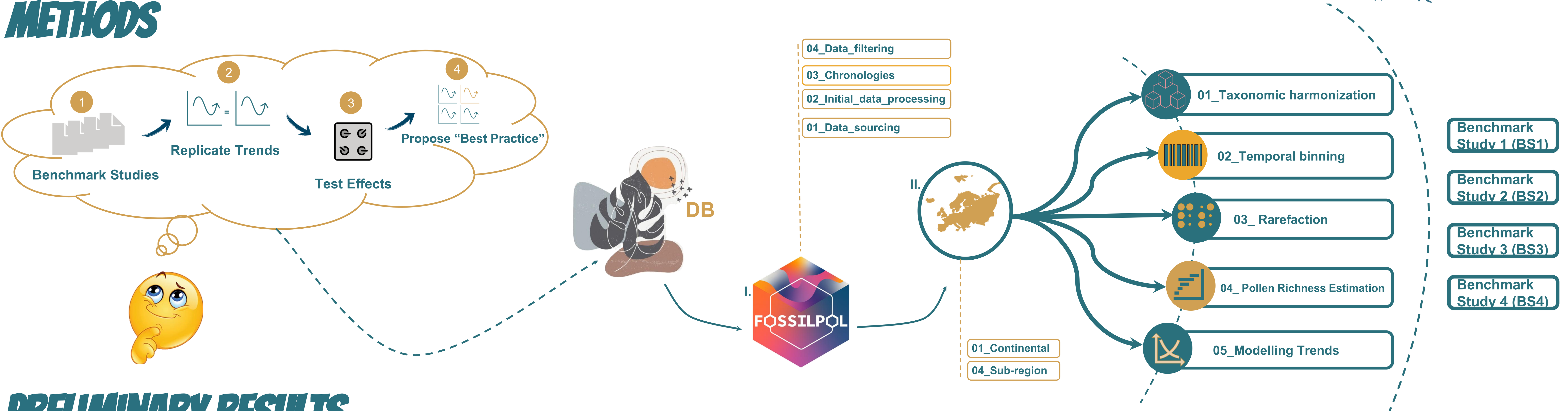
BACKGROUND

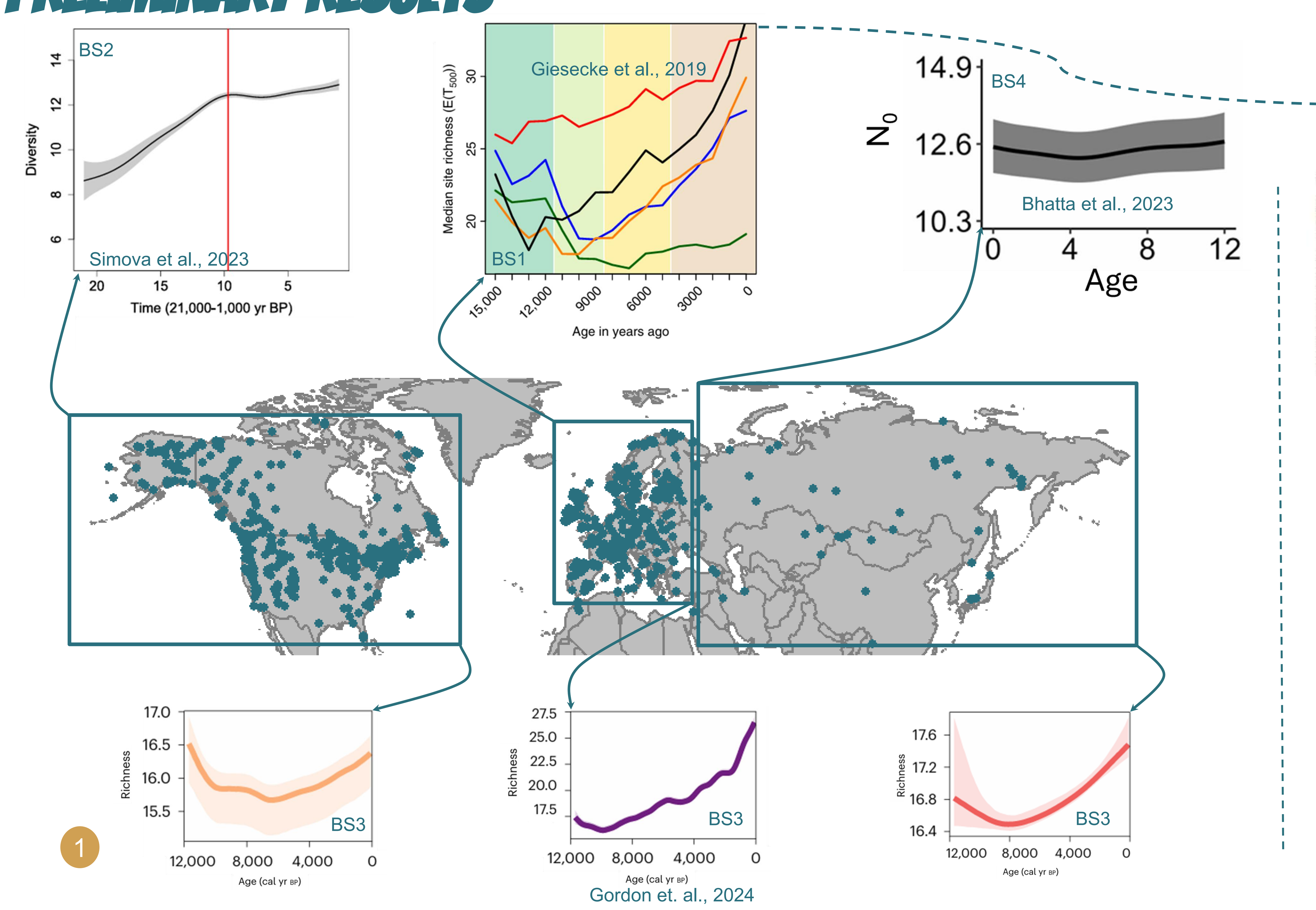
With the progress and advances of quantitative methods in paleoecology, the estimation of vegetation diversity through the analyses of fossil pollen has significantly improved. However, differences in data processing, method of diversity estimation, and usage of statistical modelling to summarise the temporal patterns on a continental scale can produce different results. To our knowledge, there has been no effort to explicitly quantify the effects of various ways of data processing, binning, richness estimation, modelling, and trend upscaling on continental pollen diversity, despite indications that these can strongly influence pollen diversity estimates.

RESEARCH QUESTION: **How do different methodologies affect the continental diversity trend across Northern hemisphere?**

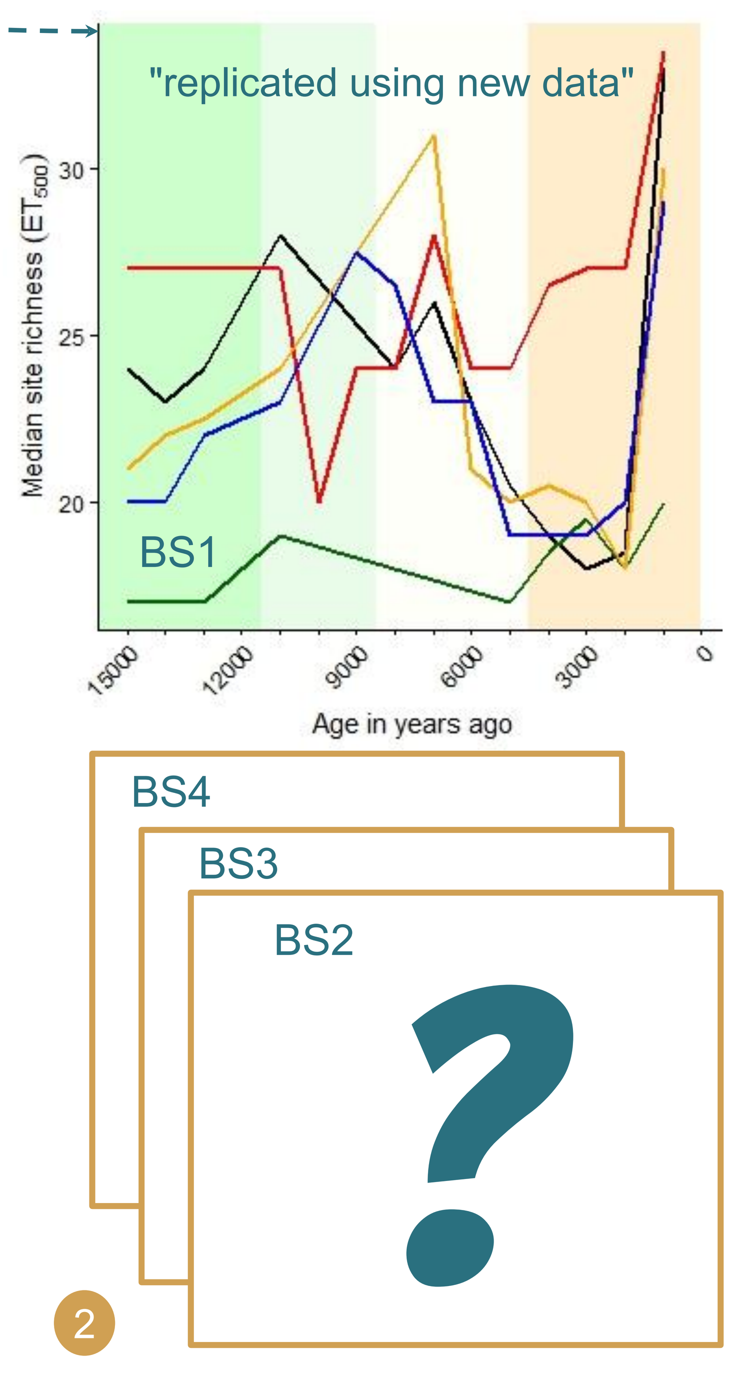
METHODS



PRELIMINARY RESULTS



REPLICATION STUDY



Continental trends of pollen diversity from benchmark studies are shown (1). Currently, the trends for Europe was already reconstructed. The methodology for BS1 was replicated as detailed as possible(i.e. age binning, rarefaction and including taxonomic harmonisation) but used more recent (high quality) data ("see map"). The study replication suggests trend instability across temporal scales which warrants further investigation. Meanwhile, floristic diversity trends from other benchmark studies (BS2 to 4) will also be replicated as close as possible, observing respective methodological details (2).

IMPLICATIONS AND NEXT STEPS

Replication for other studies will be done mainly alongside a battery of new analyses with slight modification of the methods to observe how severely it change the observed continental diversity trends.

A substantial methodological R codebase was developed and written in a functional-programming style to support subsequent numerical analyses.

Insights from replication study can lead to an improved interpretation of past vegetation diversity, which may serve as a baseline for predicting diversity amidst global environmental changes.

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Project Information @ LoQE Lab Webpage

Bryan-Novio-Holocene-Diversity-Project-Dataset

Charles University

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